Elementary Analysis

Chapter 1, Section 4

The Completeness Axiom
Exercises 4.1–4.16, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 1. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, letters in Section 3 denote real numbers. The empty sum is 0, and $0! = 1$.

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4 The completeness axiom

4.1 Exercise 4.1: Three upper bounds and a common analysis of the sets

What we must show. For each of the twenty-three sets in parts (a)–(w), give three distinct real upper bounds when such bounds exist. Otherwise, give the requested answer "NBA," meaning "not bounded above."

Plan. Exercises 4.2, 4.3, and 4.4 ask about the very same sets. We first analyze each set once, including the sharpness of its endpoint bounds. Four answer lists then state the answer to every part of every exercise and link it to its proof. This keeps the proofs of the intersections, the prime-number assertion, and the trigonometric values available without repeating them four times.

Definitions and tools for all four exercises

An *upper bound* $U \in \mathbb{R}$ for a set S must satisfy $s \le U$ for every $s \in S$. A *lower bound* $L \in \mathbb{R}$ must satisfy $L \le s$ for every $s \in S$. Thus, once U is known to be an upper bound, every number at least U is another upper bound: if $s \le U$ and $U \le V$, then $s \le V$. Similarly, every number at most a known lower bound is another lower bound.

For a proposed supremum u, we prove two things:

1. $s \le u$ for every member s of the set;

2. for each real $U < u$, there is a member $s > U$.

The second condition means that no number smaller than u is an upper bound, so u is the *least* upper bound. For a proposed infimum ℓ, the corresponding conditions are that $\ell \le s$ for every member, and that every real $L > \ell$ has some member $s < L$. If a bound belongs to the set, that member itself proves sharpness: any smaller upper bound, or larger lower bound, fails on that member. Thus a maximum is a supremum, and a minimum is an infimum.

All suprema and infima in these four exercises are *real* numbers. An unbounded-above set has "NO sup," and an unbounded-below set has "NO inf," in the conventions of this section. These answers are not assignments of extended-real values.

Tool 1: The Archimedean property and small reciprocals.

The Archimedean property implies that, for every real B, there is $n \in \mathbb{N}$ with $n > B$. If $B \le 0$, $n = 1$ works; if $B > 0$, apply the Archimedean property with $a = 1$ and $b = B$. Consequently, for every $\delta > 0$, there is an $n \in \mathbb{N}$ such that

$$n > \frac{1}{\delta} \quad \text{and therefore} \quad 0 < \frac{1}{n} < \delta.$$

To justify the last strict inequality, multiply $n > 1/\delta$ by the positive number δ, obtaining $n\delta > 1$, and then divide by the positive integer n.

Tool 2: Rational points in an open interval make its endpoints sharp.

Let $c, d \in \mathbb{R}$ with $c < d$, and consider

$$T = \{r \in \mathbb{Q} : c < r < d\}.$$

The density of $\mathbb{Q}$ shows that T is nonempty. Its defining inequalities give the upper bound d and the lower bound c. If $U < d$, put $v = \max\{U, c\}$. Both U and c are less than d, so $v < d$. Density provides a rational r with

$$\max\{U, c\} < r < d.$$

Then $c < r < d$, so $r \in T$, and $r > U$, so U is not an upper bound. This proves $\sup T = d$.

If $L > c$, put $w = \min\{L, d\}$. Both L and d are greater than c, so $c < w$. Density provides a rational r with

$$c < r < \min\{L, d\}.$$

This r belongs to T and satisfies $r < L$, proving that L is not a lower bound. Hence $\inf T = c$. Neither endpoint belongs to T, since membership requires strict inequalities. The same witnesses also prove that the real interval (c, d) has supremum d and infimum c, because those rational witnesses are real members of it.

Tool 3: Converting a quadratic inequality into an interval.

For $c > 0$ and $x \in \mathbb{R}$,

$$x^2 < c^2 \quad \Longleftrightarrow \quad -c < x < c.$$

For the reverse direction, if $-c < x < c$, then $c + x > 0$ and $c - x > 0$. Their product is positive, so

$$0 < (c - x)(c + x) = c^2 - x^2,$$

which says $x^2 < c^2$. For the forward direction, if $x \geq c > 0$, then $x^2 \geq c^2$, contradicting the strict inequality. If $x \leq -c$, then $-x \geq c > 0$ and $x^2 = (-x)^2 \geq c^2$, again a contradiction. Thus neither $x \geq c$ nor $x \leq -c$ is possible, proving $-c < x < c$.

Tool 4: Named numerical bounds used for the familiar constants.

We use the familiar estimates

$$3 < \pi < \frac{22}{7}, \qquad 2 < e < 3.$$

In particular $2 < e < 3 < \pi < 4$, because $22/7 < 4$ follows from $22 < 28$. These estimates suffice to order the finite sets involving π and e; decimal approximations are unnecessary.

A proof for every set (a)–(w)

Analysis (a): The closed interval $[0, 1]$.

Every member s satisfies $0 \leq s \leq 1$, by the definition of this interval. Both 0 and 1 belong to it. Thus 0 is its minimum and infimum, and 1 is its maximum and supremum. In particular $1, 2, 3$ are upper bounds and $0, -1, -2$ are lower bounds.

Analysis (b): The open interval $(0, 1)$.

Every member satisfies $0 < s < 1$, so 0 is a lower bound and 1 is an upper bound. Tool 2 with $c = 0$, $d = 1$ proves that no upper bound smaller than 1, and no lower bound larger than 0, can work. Therefore $\sup(0, 1) = 1$ and $\inf(0, 1) = 0$. Neither endpoint is a member, so neither extremal bound is attained. Upper bounds include $1, 2, 3$; lower bounds include $0, -1, -2$.

Analysis (c): The finite set $\{2, 7\}$.

Its two members satisfy $2 \leq 2 \leq 7$ and $2 \leq 7 \leq 7$. Consequently 2 bounds every member from below and 7 bounds every member from above. Each is itself in the set. Hence the minimum and infimum are 2, while the maximum and supremum are 7. Examples of upper bounds are $7, 8, 9$; examples of lower bounds are $2, 1, 0$.

Analysis (d): The finite set $\{\pi, e\}$.

Tool 4 gives $2 < e < 3 < \pi < 4$, and therefore $e < \pi$. Thus every member of this two-element set is at least e and at most π. Both endpoints are members, so its infimum is e and its supremum is π, both attained. Since every member is less than 4, the three distinct numbers $4, 5, 6$ are upper bounds. Since every member is greater than 2, the distinct numbers $2, 1, 0$ are lower bounds.

Analysis (e): The reciprocal set $\{1/n : n \in \mathbb{N}\}$.

For $n \geq 1$, positivity and $n \geq 1$ give

$$0 < \frac{1}{n} \leq 1.$$

At $n = 1$, the value is 1, so 1 is the maximum and supremum. The number 0 is a lower bound, but it is not a member since a positive reciprocal cannot equal 0. If $L > 0$, Tool 1 supplies n with $1/n < L$. This member defeats L as a lower bound, so 0 is the greatest lower bound. Thus the infimum is 0 and is not attained. Upper bounds include $1, 2, 3$; lower bounds include $0, -1, -2$.

Analysis (f): The singleton $\{0\}$.

Its only member is 0. Therefore 0 is both a lower and an upper bound, and is both the minimum and the maximum. It follows that the supremum and infimum both equal 0. The numbers $0, 1, 2$ are three distinct upper bounds, and $0, -1, -2$ are three distinct lower bounds.

Analysis (g): The union $[0, 1] \cup [2, 3]$.

A member s belongs to at least one of the two intervals. If $s \in [0, 1]$, then $0 \leq s \leq 1 \leq 3$. If $s \in [2, 3]$, then $0 \leq 2 \leq s \leq 3$. Thus every member satisfies $0 \leq s \leq 3$. The member 0 comes from the first interval, and the member 3 comes from the second. Hence the infimum is the attained minimum 0, and the supremum is the attained maximum 3. The gap between the two intervals changes neither endpoint bound. Upper bounds include $3, 4, 5$ and lower bounds include $0, -1, -2$.

Analysis (h): The infinite union $\bigcup_{n=1}^{\infty}[2n, 2n + 1]$.

Membership in the union means membership in *at least one* interval. If s belongs to it, there is an $n \in \mathbb{N}$ with $2n \leq s \leq 2n + 1$. Because $n \geq 1$, we obtain $2 \leq 2n \leq s$. The number 2 belongs to the interval at $n = 1$, namely $[2, 3]$. Thus 2 is the minimum and infimum, and $2, 1, 0$ are lower bounds.

There is no upper bound. To see this for an arbitrary proposed $M \in \mathbb{R}$, use Tool 1 to choose

$$n > \max\{1, M/2\}.$$

The member $2n \in [2n, 2n + 1]$ satisfies $2n > M$. Thus every proposed real upper bound fails. This set is NBA and has NO sup.

Analysis (i): The intersection $\bigcap_{n=1}^{\infty}[-1/n, 1 + 1/n]$.

Call this set I. We prove the exact identity $I = [0, 1]$ by both inclusions. If $0 \leq x \leq 1$, then for every $n \in \mathbb{N}$,

$$-\frac{1}{n} < 0 \leq x \leq 1 < 1 + \frac{1}{n}.$$

Thus x belongs to every displayed closed interval, proving $[0,1] \subseteq I$.

For the other inclusion, suppose $x \notin [0,1]$. If $x < 0$, let $\delta = -x > 0$. Tool 1 gives an n with $1/n < \delta = -x$. Multiplying this inequality by -1 reverses its direction:

$$-\frac{1}{n} > x.$$

Therefore x fails the lower-endpoint requirement in that particular interval, so $x \notin I$. If $x > 1$, let $\delta = x - 1 > 0$ and choose n with $1/n < x - 1$. Then $1 + 1/n < x$, so x fails the upper-endpoint requirement and again is not in I. These two cases exclude every real number outside $[0,1]$, proving $I \subseteq [0,1]$.

It follows that $I = [0,1]$. Its infimum is the member 0 and its supremum is the member 1. Upper bounds are $1, 2, 3$, for example, and lower bounds are $0, -1, -2$.

Analysis (j): The set $\{1 - 1/3^n : n \in \mathbb{N}\}$.

The denominator is 3^n, so the expression is $1 - 3^{-n}$. For $n \geq 1$, $3^n \geq 3$, and hence

$$0 < \frac{1}{3^n} \leq \frac{1}{3}, \qquad \frac{2}{3} = 1 - \frac{1}{3} \leq 1 - \frac{1}{3^n} < 1.$$

At $n = 1$, the lower value $2/3$ occurs. Thus $2/3$ is the minimum and infimum. The number 1 is an upper bound but is not a member.

To prove that this upper bound is sharp without assuming convergence, we first verify $3^n \geq n$ for $n \in \mathbb{N}$. At $n = 1$, $3 \geq 1$. If $3^n \geq n$, multiplication by $3 > 0$ gives $3^{n+1} \geq 3n$. Also $3n - (n+1) = 2n - 1 \geq 1 > 0$, so $3n \geq n + 1$. Induction proves the auxiliary inequality.

Now fix a proposed smaller bound $U < 1$ and set $\delta = 1 - U > 0$. Choose n by Tool 1 so that $1/n < \delta$. Since $3^n \geq n > 0$,

$$\frac{1}{3^n} \leq \frac{1}{n} < 1 - U, \qquad 1 - \frac{1}{3^n} > U.$$

This member exceeds U, so U is not an upper bound. Therefore the supremum is 1, not attained. Upper bounds include $1, 2, 3$; lower bounds include $2/3, 0, -1$.

Analysis (k): The set $\{n + (-1)^n/n : n \in \mathbb{N}\}$.

The numerator $(-1)^n$ equals either 1 or -1. Therefore, for each $n \in \mathbb{N}$,

$$n + \frac{(-1)^n}{n} \geq n - \frac{1}{n} \geq n - 1 \geq 0.$$

Here the first inequality uses $(-1)^n \geq -1$ and division by $n > 0$; the second uses $1/n \leq 1$; the third uses $n \geq 1$. At $n = 1$, the expression is $1 + (-1)/1 = 0$. Thus 0 is the minimum and infimum, and $0, -1, -2$ are lower bounds.

For any proposed real upper bound M, choose $n \in \mathbb{N}$ with $n > M + 1$, using Tool 1. The same estimate gives

$$n + \frac{(-1)^n}{n} \geq n - 1 > M.$$

Thus the set is NBA and has NO sup. The oscillating correction $(-1)^n/n$ does not keep the growing term n bounded.

Analysis (l): The rational half-line $\{r \in \mathbb{Q} : r < 2\}$.

All members are less than 2, so 2 is an upper bound. If $U < 2$, density provides a rational r with $U < r < 2$. That r is a member and is larger than U, so no $U < 2$ is an upper bound. Hence the supremum is 2. It is not attained because $r = 2$ fails the strict membership inequality. Upper bounds include $2, 3, 4$.

There is no lower bound: for a proposed $L \in \mathbb{R}$, choose $n \in \mathbb{N}$ with $n > \max\{1, -L\}$. Then $-n$ is an integer, hence rational, and $-n < 0 < 2$, so $-n$ is in the set. Also $n > -L$ gives $-n < L$. Thus every proposed lower bound fails. The answers for lower boundedness and infimum are NBB and NO inf.

Analysis (m): The rational set $\{r \in \mathbb{Q} : r^2 < 4\}$.

Tool 3 with $c = 2$ gives

$$\{r \in \mathbb{Q} : r^2 < 4\} = \{r \in \mathbb{Q} : -2 < r < 2\}.$$

Thus -2 is a lower bound and 2 is an upper bound. To see sharpness explicitly, for $U < 2$ density supplies a rational $\max\{U, -2\} < r < 2$, a member exceeding U. For $L > -2$ it supplies a rational $-2 < r < \min\{L, 2\}$, a member below L. These are exactly the two endpoint arguments of Tool 2. Thus the supremum is 2 and the infimum is -2. Neither is a member: $(2)^2 = (-2)^2 = 4$, whereas membership requires a square strictly less than 4. Three upper bounds are $2, 3, 4$; three lower bounds are $-2, -3, -4$.

Analysis (n): The rational set $\{r \in \mathbb{Q} : r^2 < 2\}$.

Let $c = \sqrt{2}$, the positive real number with $c^2 = 2$. Tool 3 gives

$$\{r \in \mathbb{Q} : r^2 < 2\} = \{r \in \mathbb{Q} : -\sqrt{2} < r < \sqrt{2}\}.$$

For $U < \sqrt{2}$, density gives a rational $\max\{U, -\sqrt{2}\} < r < \sqrt{2}$; it is a member above U. For $L > -\sqrt{2}$, density gives a rational $-\sqrt{2} < r < \min\{L, \sqrt{2}\}$; it is a member below L. Together with the bounding inequalities, this proves

$$\sup S = \sqrt{2}, \qquad \inf S = -\sqrt{2}.$$

Neither endpoint belongs to the set, since its square equals 2 instead of being less than 2. This exclusion already follows from the strict inequality; it does not require a separate irrationality argument.

Since $0 < \sqrt{2} < 2$, every member lies strictly between -2 and 2. For the last comparison, $\sqrt{2} \geq 2$ would imply $2 = (\sqrt{2})^2 \geq 4$, a contradiction. Therefore $2, 3, 4$ are upper bounds and $-2, -3, -4$ are lower bounds.

Analysis (o): The negative half-line $\{x \in \mathbb{R} : x < 0\}$.

The number 0 is an upper bound. If $U < 0$, then

$$U < \frac{U}{2} < 0,$$

because $U/2 - U = -U/2 > 0$ and $U/2$ is still negative. Thus $U/2$ is a member above the proposed bound U, proving that the supremum is 0. The value 0 is not a member. Upper bounds include $0, 1, 2$.

For an arbitrary proposed lower bound $L \in \mathbb{R}$, take $x = \min\{L, 0\} - 1$. Then

$$x < \min\{L, 0\} \leq L, \qquad x < \min\{L, 0\} \leq 0.$$

Hence x is a negative real number below L. This proves NBB and therefore NO inf.

Analysis (p): The finite set $\{1, \pi/3, \pi^2, 10\}$.

The third entry is π^2, not a cube. We verify the ordering using Tool 4. Dividing $3 < \pi < 22/7$ by the positive number 3 gives

$$1 < \frac{\pi}{3} < \frac{22}{21} < 2,$$

where $22/21 < 2$ follows from $22 < 42$. Squaring the positive inequalities $3 < \pi < 22/7$ gives

$$9 < \pi^2 < \frac{484}{49} < 10.$$

The final strict inequality holds since $484 < 490 = 10 \cdot 49$. Combining these comparisons with $2 < 9$, we obtain

$$1 < \frac{\pi}{3} < 2 < 9 < \pi^2 < 10.$$

Thus 1 is the member of smallest value, and 10 is the member of largest value. The infimum is the attained minimum 1, and the supremum is the attained maximum 10. Upper bounds include $10, 11, 12$; lower bounds include $1, 0, -1$.

Analysis (q): The finite set $\{0, 1, 2, 4, 8, 16\}$.

The displayed members satisfy

$$0 < 1 < 2 < 4 < 8 < 16.$$

Every member is at least 0 and at most 16, and both 0 and 16 appear in the set. Hence the minimum and infimum equal 0, and the maximum and supremum equal 16. Upper bounds include $16, 17, 18$, while lower bounds include $0, -1, -2$.

Analysis (r): The intersection $\bigcap_{n=1}^{\infty}(1 - 1/n, 1 + 1/n)$.

Let J denote this intersection. For each $n \in \mathbb{N}$, $1/n > 0$, so $1 - 1/n < 1 < 1 + 1/n$. Therefore 1 belongs to every one of the open intervals and hence to J.

Now suppose $x \neq 1$. If $x < 1$, take $\delta = 1 - x > 0$ and use Tool 1 to choose n with $1/n < 1 - x$. Rearranging gives $x < 1 - 1/n$. Thus x lies below the lower endpoint of that particular open interval and cannot belong to J. If $x > 1$, choose n with $1/n < x - 1$. Then $1 + 1/n < x$, and x lies above the upper endpoint of that interval, so again $x \notin J$. These two cases exclude every $x \neq 1$.

It follows that $J = \{1\}$. Thus both its infimum and its supremum are 1, and both are attained. The fact that the intersected intervals are open does not prevent their intersection from being a singleton. Three upper bounds are $1, 2, 3$; three lower bounds are $1, 0, -1$.

Analysis (s): Reciprocals of positive primes.

The set is

$$P = \{1/p : p \in \mathbb{N} \text{ and } p \text{ is prime}\}.$$

Every positive prime p satisfies $p \geq 2$, so $0 < 1/p \leq 1/2$. The integer 2 is prime: its only positive integer factors are 1 and 2. Consequently $1/2$ belongs to P, and is its maximum and supremum. The number 0 is a lower bound and is not a member. To show it is the *greatest* lower bound, we need primes as large as we wish.

The needed prime-number fact, with proof. First, each integer $m \geq 2$ has a prime divisor. The finite nonempty set

$$D = \{d \in \{2, 3, \ldots, m\} : d \text{ divides } m\}$$

contains m, so it has a smallest member p. If p were not prime, it would have a positive divisor a other than 1 and p, with $2 \leq a < p$. Writing $p = ab$ and $m = pc$ with integers b, c gives $m = a(bc)$, so a would divide m and belong to D. That contradicts the minimality of p. Hence p is prime.

Now let $B \in \mathbb{R}$ be arbitrary. By Tool 1 choose an integer $N > \max\{B, 2\}$. The integer $m = N! + 1$ is at least 2, so it has a prime divisor p. If $p \leq N$, then p is one of the factors in the product $N! = 1 \cdot 2 \cdots N$. Thus both $N!$ and $N! + 1$ would be divisible by p. Subtracting their integer-multiple representations would make

$$(N! + 1) - N! = 1$$

divisible by p. This is impossible for an integer $p \geq 2$: if $1 = pq$ for an integer q, then $q \neq 0$ and $1 = p\,|q| \geq p \geq 2$ after taking absolute values, a contradiction. Therefore $p > N > B$. This proves that the primes are unbounded above.

Finally let $L > 0$ be an arbitrary proposed lower bound for P. Apply the fact just proved with $B = 1/L$ to obtain a prime $p > 1/L$. Multiplication by $L > 0$ gives $pL > 1$, and division by $p > 0$ then gives $1/p < L$. This member of P defeats the proposed lower bound L. Therefore

$$\inf P = 0, \qquad \sup P = \frac{1}{2}.$$

Only the supremum is attained. Three upper bounds are $1/2, 1, 2$; three lower bounds are $0, -1, -2$.

Analysis (t): The real set $\{x \in \mathbb{R} : x^3 < 8\}$.

We first identify the inequality's exact solution set. Multiplication shows

$$\begin{aligned}(x - 2)(x^2 + 2x + 4) &= x^3 + 2x^2 + 4x - 2x^2 - 4x - 8 \\ &= x^3 - 8.\end{aligned}$$

The second factor is always positive, because

$$x^2 + 2x + 4 = (x + 1)^2 + 3 \geq 3 > 0.$$

The sign of $x^3 - 8$ is therefore the sign of $x - 2$. It follows that $x^3 < 8$ if and only if $x < 2$.

Thus 2 is an upper bound. If $U < 2$, the midpoint $x = (U + 2)/2$ satisfies

$$x - U = \frac{2 - U}{2} > 0, \qquad 2 - x = \frac{2 - U}{2} > 0.$$

Hence $U < x < 2$, so x belongs to the set and is above U. This proves that the supremum is 2. It is not attained since $2^3 = 8$ fails the strict inequality. Upper bounds include $2, 3, 4$.

For any $L \in \mathbb{R}$, let $x = \min\{L, 2\} - 1$. Then $x < L$ and $x < 2$, so x is a member below the proposed lower bound L. Thus the set is NBB and has NO inf.

Analysis (u): The set of real squares $\{x^2 : x \in \mathbb{R}\}$.

Every square is nonnegative, so 0 is a lower bound. Taking $x = 0$ gives the member $0^2 = 0$. Therefore 0 is the minimum and infimum, and $0, -1, -2$ are lower bounds.

To defeat a proposed upper bound $M \in \mathbb{R}$, use Tool 1 to choose $n \in \mathbb{N}$ with $n > \max\{M, 1\}$. Since $n \geq 1$, $n^2 \geq n > M$. Taking $x = n$ produces the member $x^2 = n^2$ above M. Thus this set is NBA and has NO sup.

Analysis (v): The values of $\cos(n\pi/3)$.

We record a period table, which also supplies the proof for (w). For a positive integer n, division of $n - 1$ by 6 gives $n = 6q + r$ with $q \geq 0$ an integer and $r \in \{1, 2, 3, 4, 5, 6\}$. Therefore

$$\frac{n\pi}{3} = 2q\pi + \frac{r\pi}{3}.$$

The period 2π of sine and cosine shows that their values depend only on r. The familiar unit-circle values are

r	$r\pi/3$	$\cos(r\pi/3)$	$\sin(r\pi/3)$
1	$\pi/3$	$1/2$	$\sqrt{3}/2$
2	$2\pi/3$	$-1/2$	$\sqrt{3}/2$
3	π	-1	0
4	$4\pi/3$	$-1/2$	$-\sqrt{3}/2$
5	$5\pi/3$	$1/2$	$-\sqrt{3}/2$
6	2π	1	0

Every positive integer has one of these residues, so no other values occur. Every row occurs at $n = r$, so none of the listed values is missing from the corresponding set.

In particular the cosine set is exactly

$$\{-1, -1/2, 1/2, 1\}.$$

Its minimum and infimum are -1, attained at $n = 3$, and its maximum and supremum are 1, attained at $n = 6$. Upper bounds include $1, 2, 3$; lower bounds include $-1, -2, -3$.

Analysis (w): The values of $\sin(n\pi/3)$.

The period argument and all six sine values were displayed in Analysis (v). They show that the sine set is exactly

$$\{-\sqrt{3}/2, 0, \sqrt{3}/2\}.$$

Because $\sqrt{3} > 0$, these three numbers have the displayed increasing order. Thus its infimum is the member $-\sqrt{3}/2$ at $n = 4$, and its supremum is the member $\sqrt{3}/2$ at $n = 1$. Also $\sqrt{3} < 2$: the contrary inequality would imply $3 = (\sqrt{3})^2 \geq 4$. Hence $0 < \sqrt{3}/2 < 1$, so all members are between -1 and 1. Three upper bounds are $1, 2, 3$; three lower bounds are $-1, -2, -3$.

Exercise 4.1: Explicit answers for all twenty-three parts

Each row gives three *distinct* upper bounds, or NBA. A number equal to the supremum is permitted as an upper bound whether or not it belongs to the set. In a bounded row, the reason proves that the smallest listed number is an upper bound; the larger two then follow by transitivity of $\leq$. The linked analysis supplies the full membership or unboundedness argument.

Part	Three upper bounds / NBA	Why these are valid
(a)	$1, 2, 3$	Every member is at most 1; Analysis (a).
(b)	$1, 2, 3$	Every member is less than 1; Analysis (b).
(c)	$7, 8, 9$	Both members are at most 7; Analysis (c).
(d)	$4, 5, 6$	$e < 3 < \pi < 4$; Analysis (d).
(e)	$1, 2, 3$	$1/n \leq 1$ for every $n \geq 1$; Analysis (e).
(f)	$0, 1, 2$	The only member is 0; Analysis (f).
(g)	$3, 4, 5$	Each interval lies at or below 3; Analysis (g).
(h)	NBA	The members $2n$ exceed any proposed bound; Analysis (h).
(i)	$1, 2, 3$	The intersection equals $[0, 1]$; Analysis (i).
(j)	$1, 2, 3$	$1 - 3^{-n} < 1$; Analysis (j).
(k)	NBA	Members are at least $n - 1$, unbounded above; Analysis (k).
(l)	$2, 3, 4$	Membership requires $r < 2$; Analysis (l).

Part	Three upper bounds / NBA	Why these are valid
(m)	$2, 3, 4$	$r^2 < 4$ implies $r < 2$; Analysis (m).
(n)	$2, 3, 4$	$r < \sqrt{2} < 2$; Analysis (n).
(o)	$0, 1, 2$	Membership requires $x < 0$; Analysis (o).
(p)	$10, 11, 12$	10 is the largest of the four members; Analysis (p).
(q)	$16, 17, 18$	16 is the largest listed member; Analysis (q).
(r)	$1, 2, 3$	The intersection is $\{1\}$; Analysis (r).
(s)	$1/2, 1, 2$	Every prime is at least 2; Analysis (s).
(t)	$2, 3, 4$	$x^3 < 8$ is equivalent to $x < 2$; Analysis (t).
(u)	NBA	Squares n^2 exceed any real bound; Analysis (u).
(v)	$1, 2, 3$	The four cosine values are at most 1; Analysis (v).
(w)	$1, 2, 3$	The largest sine value is $\sqrt{3}/2 < 1$; Analysis (w).

Conclusion. The only sets in Exercise 4.1 that are not bounded above are (h), (k), and (u). Every other part has the three distinct upper bounds explicitly listed and justified above.

4.2 Exercise 4.2: Three lower bounds for each of the same sets

What we must show. For each set (a)–(w) from Exercise 4.1, list three distinct real lower bounds or give NBB, meaning "not bounded below."

Plan. Use the lower-endpoint conclusions of the shared analyses. If ℓ is a lower bound and $L \leq \ell$, then $L \leq \ell \leq s$ for each member s, so L is also a lower bound. Consequently it suffices in each bounded row to justify the largest of the three listed candidates; the other two are smaller. For an NBB row, the linked proof constructs, for every $L \in \mathbb{R}$, a member strictly below L.

Part	Three lower bounds / NBB	Why these are valid
(a)	$0, -1, -2$	Every member is at least 0; Analysis (a).
(b)	$0, -1, -2$	Every member is positive; Analysis (b).
(c)	$2, 1, 0$	Both members are at least 2; Analysis (c).
(d)	$2, 1, 0$	$2 < e < \pi$; Analysis (d).
(e)	$0, -1, -2$	Every $1/n$ is positive; Analysis (e).
(f)	$0, -1, -2$	The only member is 0; Analysis (f).
(g)	$0, -1, -2$	Both intervals lie at or above 0; Analysis (g).
(h)	$2, 1, 0$	Each interval starts at $2n \geq 2$; Analysis (h).
(i)	$0, -1, -2$	The intersection is $[0, 1]$; Analysis (i).
(j)	$2/3, 0, -1$	$1 - 3^{-n} \geq 2/3$; Analysis (j).
(k)	$0, -1, -2$	$n + (-1)^n/n \geq 0$; Analysis (k).
(l)	NBB	Negative integers in the set go below any L; Analysis (l).

Part	Three lower bounds / NBB	Why these are valid
(m)	$-2, -3, -4$	$r^2 < 4$ implies $r > -2$; Analysis (m).
(n)	$-2, -3, -4$	$r > -\sqrt{2} > -2$; Analysis (n).
(o)	NBB	$\min\{L, 0\} - 1$ is a member below L; Analysis (o).
(p)	$1, 0, -1$	1 is the smallest of the four members; Analysis (p).
(q)	$0, -1, -2$	0 is the smallest listed member; Analysis (q).
(r)	$1, 0, -1$	The only member of the intersection is 1; Analysis (r).
(s)	$0, -1, -2$	All prime reciprocals are positive; Analysis (s).
(t)	NBB	$\min\{L, 2\} - 1$ satisfies $x^3 < 8$ and $x < L$; Analysis (t).
(u)	$0, -1, -2$	Every real square is nonnegative; Analysis (u).
(v)	$-1, -2, -3$	The cosine values are at least -1; Analysis (v).
(w)	$-1, -2, -3$	The least sine value is $-\sqrt{3}/2 > -1$; Analysis (w).

Conclusion. The only sets in Exercise 4.2 that are not bounded below are (l), (o), and (t). Each remaining part has three distinct lower bounds in its row.

4.3 Exercise 4.3: The real supremum of each set

What we must show. For each of the twenty-three sets, give its supremum when that real least upper bound exists; otherwise write NO sup.

Plan. A mere upper bound is insufficient: we also need leastness. The shared analyses prove leastness either by exhibiting the proposed bound as a member or by constructing members above every smaller candidate. The word "attained" in the table means the supremum belongs to the set, in which case it is also the maximum. Failure of attainment does not prevent a supremum from existing.

Part	Supremum	Bound and leastness proof
(a)	1	An upper bound and a member; attained. Analysis (a).
(b)	1	Upper endpoint is sharp by Tool 2; not attained. Analysis (b).
(c)	7	The larger member is an upper bound; attained. Analysis (c).
(d)	π	$e < \pi$ and π is a member; attained. Analysis (d).
(e)	1	$1/n \leq 1$, with equality at $n = 1$; attained. Analysis (e).
(f)	0	The sole member is its upper bound; attained. Analysis (f).
(g)	3	All members are at most 3 and 3 is a member; attained. Analysis (g).
(h)	NO sup	There is no real upper bound at all. Analysis (h).
(i)	1	The intersection equals $[0, 1]$; attained. Analysis (i).
(j)	1	Members exceed any $U < 1$ but never equal 1; not attained. Analysis (j).
(k)	NO sup	The members exceed every real upper-bound candidate. Analysis (k).
(l)	2	Density gives a rational member above any $U < 2$; not attained. Analysis (l).

Part	Supremum	Bound and leastness proof
(m)	2	The rational interval has sharp upper endpoint 2; not attained. Analysis (m).
(n)	$\sqrt{2}$	Density proves the upper endpoint is sharp; not attained. Analysis (n).
(o)	0	For $U < 0$, the member $U/2$ is above U; not attained. Analysis (o).
(p)	10	All four values are at most the member 10; attained. Analysis (p).
(q)	16	The largest listed member is 16; attained. Analysis (q).
(r)	1	The intersection is the singleton $\{1\}$; attained. Analysis (r).
(s)	1/2	Every prime is at least 2, and 2 is prime; attained. Analysis (s).
(t)	2	The set is $\{x : x < 2\}$, with members above every $U < 2$; not attained. Analysis (t).
(u)	NO sup	Real squares exceed every proposed upper bound. Analysis (u).
(v)	1	The period table has maximum 1, occurring at $n = 6$; attained. Analysis (v).
(w)	$\sqrt{3}/2$	The period table has this largest value at $n = 1$; attained. Analysis (w).

Conclusion. Parts (h), (k), and (u) have NO sup in $\mathbb{R}$. Every other supremum is the real number stated in its row, with its upper-bound and leastness requirements proved in the linked analysis.

4.4 Exercise 4.4: The real infimum of each set

What we must show. Give the real greatest lower bound of each set (a)–(w), or write NO inf if the set has no real infimum.

Plan. Reverse the roles of upper and lower bounds in Exercise 4.3. For an attained lower bound, the member itself defeats every larger candidate. For a nonattained infimum, the linked analysis constructs members below every larger candidate. "NO inf" is used exactly where there is no real lower bound.

Part	Infimum	Bound and greatestness proof
(a)	0	A lower bound and a member; attained. Analysis (a).
(b)	0	The open interval has sharp lower endpoint 0; not attained. Analysis (b).
(c)	2	The smaller member is a lower bound; attained. Analysis (c).
(d)	e	$e < \pi$ and e is a member; attained. Analysis (d).
(e)	0	Positive reciprocals lie below any $L > 0$; not attained. Analysis (e).
(f)	0	The sole member is its lower bound; attained. Analysis (f).
(g)	0	Every member is at least the member 0; attained. Analysis (g).
(h)	2	Every interval starts at least at 2, and 2 occurs; attained. Analysis (h).
(i)	0	The intersection is $[0, 1]$; attained. Analysis (i).
(j)	2/3	Every value is at least 2/3, with equality at $n = 1$; attained. Analysis (j).
(k)	0	Every value is nonnegative, with equality at $n = 1$; attained. Analysis (k).
(l)	NO inf	Negative integer members go below any proposed lower bound. Analysis (l).

Part	Infimum	Bound and greatestness proof
(m)	-2	Density makes the strict lower endpoint sharp; not attained. Analysis (m).
(n)	$-\sqrt{2}$	Density makes the strict lower endpoint sharp; not attained. Analysis (n).
(o)	NO inf	Negative real members go below every real L. Analysis (o).
(p)	1	The other three members are greater than the member 1; attained. Analysis (p).
(q)	0	The least of the listed members is 0; attained. Analysis (q).
(r)	1	The intersection equals $\{1\}$; attained. Analysis (r).
(s)	0	Unbounded primes give a reciprocal below any $L > 0$; not attained. Analysis (s).
(t)	NO inf	The half-line $x < 2$ has members below every real L. Analysis (t).
(u)	0	Every square is nonnegative and $0^2 = 0$; attained. Analysis (u).
(v)	-1	The smallest period-table value occurs at $n = 3$; attained. Analysis (v).
(w)	$-\sqrt{3}/2$	The smallest sine value occurs at $n = 4$; attained. Analysis (w).

Conclusion. Parts (l), (o), and (t) have NO inf in $\mathbb{R}$. Every other part has the real infimum in its row. In particular, (e) and (s) have infimum 0 without containing 0, while the open-interval intersection in (r) has both supremum and infimum equal to its actual member 1.

4.5 Exercise 4.5: When a supremum is a maximum

What we must show. Let S be a nonempty subset of $\mathbb{R}$ that is bounded above. Prove that if $\sup S \in S$, then $\sup S = \max S$.

Step 1: recall the two requirements for a maximum.

A number M is the maximum of S exactly when

(i) M is an element of S;

(ii) every $s \in S$ satisfies $s \leq M$.

The first requirement concerns membership; the second concerns order.

Step 2: check those requirements for the supremum.

Since S is nonempty and bounded above, completeness supplies the real number $M = \sup S$. The hypothesis of the exercise is precisely $M \in S$, so requirement (i) holds. A supremum is an upper bound, so $s \leq M$ for every $s \in S$; requirement (ii) holds as well.

Conclusion. The number $\sup S$ belongs to S and is at least every member of S. By the definition of maximum, $\boxed{\sup S = \max S}$. The additional membership hypothesis is essential: an upper bound need not be an element of its set.

4.6 Exercise 4.6: Comparing the infimum and supremum

What we must show. For a nonempty bounded set $S \subseteq \mathbb{R}$, prove (a) $\inf S \leq \sup S$, and (b) describe S when these two numbers are equal.

Part (a), Step 1: identify the real bounds and choose an element.

Because S is nonempty and bounded, the completeness axiom and its lower-bound version give real numbers

$$m = \inf S, \qquad M = \sup S.$$

Nonemptiness permits us to choose a specific $s_0 \in S$.

Part (a), Step 2: apply the meanings of lower and upper bound.

Since m is a lower bound, $m \leq s_0$. Since M is an upper bound, $s_0 \leq M$. Transitivity of $\leq$ gives

$$m \leq s_0 \leq M \quad \Longrightarrow \quad m \leq M.$$

Conclusion. For every nonempty bounded $S \subseteq \mathbb{R}$, $\boxed{\inf S \leq \sup S}$. Choosing s_0 is why nonemptiness matters in this argument.

Part (b), Step 1: let the common value be c.

Assume $\inf S = \sup S = c$. For an arbitrary $s \in S$, the lower-bound and upper-bound properties give

$$c \leq s \leq c.$$

By antisymmetry of the order, $s = c$. Since s was arbitrary, every element of S equals c, and hence $S \subseteq \{c\}$.

Part (b), Step 2: use nonemptiness to get the reverse inclusion.

There is an $s_0 \in S$. The preceding step gives $s_0 = c$, so $c \in S$. Thus $\{c\} \subseteq S$, and the two inclusions give $S = \{c\}$. Conversely, for the singleton $\{c\}$, the element c is both its maximum and its minimum. Every upper bound must be at least c, and every lower bound must be at most c. Therefore both its supremum and its infimum equal c.

Conclusion. Equality of the infimum and supremum occurs exactly when S is a singleton: $\boxed{\inf S = \sup S = c \iff S = \{c\}}$.

4.7 Exercise 4.7: Bounds of a subset and the supremum of a union

What we must show. Let S, T be nonempty bounded subsets of $\mathbb{R}$. Prove (a) that $S \subseteq T$ implies $\inf T \leq \inf S \leq \sup S \leq \sup T$, and (b) that $\sup(S \cup T) = \max\{\sup S, \sup T\}$ without assuming either set contains the other.

Part (a), Step 1: prove the comparison between the infima.

Assume $S \subseteq T$. Since $\inf T$ is a lower bound of T, it satisfies $\inf T \leq t$ for every $t \in T$. Every $s \in S$ is also an element of T, so

$$\inf T \leq s \qquad \text{for every } s \in S.$$

Consequently $\inf T$ is a lower bound of S. The number $\inf S$ is the *greatest* lower bound of S, so every lower bound of S is at most $\inf S$. In particular,

$$\inf T \leq \inf S.$$

Part (a), Step 2: prove the middle comparison.

Exercise 4.6(a), applied to the nonempty bounded set S, gives

$$\inf S \leq \sup S.$$

Part (a), Step 3: prove the comparison between the suprema.

Because $\sup T$ is an upper bound of T, every $s \in S \subseteq T$ satisfies $s \leq \sup T$. Thus $\sup T$ is an upper bound of S. Since $\sup S$ is the *least* upper bound of S,

$$\sup S \leq \sup T.$$

Conclusion. Combining the three inequalities gives $\boxed{\inf T \leq \inf S \leq \sup S \leq \sup T}$ whenever $S \subseteq T$.

Part (b), Step 1: verify that the union has real bounds.

Write

$$U = S \cup T, \qquad L = \min\{\inf S, \inf T\}, \qquad M = \max\{\sup S, \sup T\}.$$

The numbers L, M are real. Since S is nonempty, the union U is nonempty. If $u \in U$, then $u \in S$ or $u \in T$. In the first case,

$$L \leq \inf S \leq u \leq \sup S \leq M;$$

in the second case,

$$L \leq \inf T \leq u \leq \sup T \leq M.$$

Therefore U is bounded below by L and above by M. In particular, $\sup U$ exists as a real number.

Part (b), Step 2: prove that $\sup U \leq M$.

Step 1 showed $u \leq M$ for every $u \in U$, so M is an upper bound of U. The least-upper-bound property gives $\sup U \leq M$.

Part (b), Step 3: prove the inequality in the other direction.

Since $S \subseteq U$, every member of S is at most $\sup U$. Hence $\sup U$ is an upper bound of S, and $\sup S \leq \sup U$. The same reasoning with $T \subseteq U$ gives $\sup T \leq \sup U$. The maximum M is one of these two numbers; both are at most $\sup U$. Thus $M \leq \sup U$.

Conclusion. The two comparisons $\sup U \leq M$ and $M \leq \sup U$ give

$$\boxed{\sup(S \cup T) = \max\{\sup S, \sup T\}}.$$

We used the containments $S \subseteq S \cup T$ and $T \subseteq S \cup T$, not a containment between S and T.

4.8 Exercise 4.8: Two sets ordered entirely below and above each other

What we must show. Let $S, T \subseteq \mathbb{R}$ be nonempty and suppose

$$s \leq t \qquad \text{for every } s \in S \text{ and every } t \in T.$$

Show (a) that S is bounded above and T is bounded below, (b) that $\sup S \leq \inf T$, and supply the examples requested in (c) and (d).

Part (a), Step 1: choose one member from each set.

Because both sets are nonempty, there are $s_0 \in S$ and $t_0 \in T$. Keep these two particular numbers fixed for this step. The hypothesis, with $t = t_0$, says

$$s \leq t_0 \qquad \text{for every } s \in S.$$

Thus the real number t_0 is an upper bound of S. The hypothesis, with $s = s_0$, says

$$s_0 \leq t \qquad \text{for every } t \in T.$$

Thus the real number s_0 is a lower bound of T.

Conclusion. S is bounded above, and T is bounded below. No lower bound for S, or upper bound for T, is required by this conclusion.

Part (b), Step 1: establish existence of the two numbers.

The set S is nonempty and bounded above, so the completeness axiom gives $\sup S \in \mathbb{R}$. The set T is nonempty and bounded below, so Corollary 4.5 gives $\inf T \in \mathbb{R}$.

Part (b), Step 2: turn the supremum into a lower bound of T.

Take an arbitrary $t \in T$. The hypothesis says that $s \leq t$ for every $s \in S$, so t is an upper bound of S. Since $\sup S$ is the least upper bound of S, we have $\sup S \leq t$. This is true for every $t \in T$. Therefore $\sup S$ is a lower bound of T. Its greatest lower bound is $\inf T$, so

$$\sup S \leq \inf T.$$

Conclusion. The separation of all pairs of set elements implies $\boxed{\sup S \leq \inf T}$. The proof does not assume that either extremal bound belongs to its set.

Part (c), Step 1: give sets that share an element.

Take

$$S = [0, 1], \qquad T = [1, 2].$$

Both sets are nonempty because 1 belongs to each. If $s \in S$ and $t \in T$, the interval definitions give $s \leq 1$ and $1 \leq t$; transitivity gives $s \leq t$, as required.

Part (c), Step 2: identify their intersection and bounds.

If $x \in S \cap T$, membership in S gives $x \leq 1$, and membership in T gives $1 \leq x$. Antisymmetry forces $x = 1$. Conversely, 1 belongs to both intervals, so

$$S \cap T = \{1\}.$$

The number 1 is the maximum of S and the minimum of T. Hence $\sup S = 1 = \inf T$.

Conclusion. The example $\boxed{S = [0,1],\ T = [1,2]}$ meets the pairwise order condition and has nonempty intersection $\{1\}$.

Part (d), Step 1: give disjoint sets with a common extremal bound.

Take

$$S = (0,1), \qquad T = (1,2).$$

The first contains $1/2$, and the second contains $3/2$, so both are nonempty. For $s \in S$ and $t \in T$, we have $s < 1 < t$, and hence $s \le t$. If x belonged to their intersection, we would have both $x < 1$ and $x > 1$, which is impossible. Thus $S \cap T = \varnothing$.

Part (d), Step 2: prove directly that $\sup S = 1$.

Every $s \in (0,1)$ satisfies $s < 1$, so 1 is an upper bound of S. To prove it is the least, take any real $u < 1$ and put

$$v = \max\{0, u\}, \qquad s = \frac{v+1}{2}.$$

Both 0 and u are less than 1, so $v < 1$. Since $v \ge 0$, the inequalities $v < s < 1$ imply $0 < s < 1$. For completeness,

$$s - v = \frac{1-v}{2} > 0, \qquad 1 - s = \frac{1-v}{2} > 0.$$

Thus $s \in S$ and $s > v \ge u$. Consequently u cannot be an upper bound of S. Every number less than 1 fails to be an upper bound, so $\sup S = 1$.

Part (d), Step 3: prove directly that $\inf T = 1$.

Every $t \in (1,2)$ satisfies $t > 1$, so 1 is a lower bound of T. For an arbitrary real $\ell > 1$, put

$$w = \min\{2, \ell\}, \qquad t = \frac{1+w}{2}.$$

Both 2 and ℓ exceed 1, so $w > 1$. We have

$$t - 1 = \frac{w-1}{2} > 0, \qquad w - t = \frac{w-1}{2} > 0.$$

Therefore $1 < t < w \le 2$, which gives $t \in T$. Also $t < w \le \ell$, so $t < \ell$. Thus ℓ cannot be a lower bound of T. Every number greater than 1 fails to be a lower bound, so $\inf T = 1$.

Conclusion. The example $\boxed{S = (0,1),\ T = (1,2)}$ has $\boxed{\sup S = \inf T = 1}$ and $\boxed{S \cap T = \varnothing}$. Equality of the bounds does not force either set to contain the common bound.

4.9 Exercise 4.9: Obtain an infimum by negating a supremum

What we must show. Complete the proof of Corollary 4.5. For a nonempty set $S \subseteq \mathbb{R}$ that is bounded below, put

$$-S = \{-s : s \in S\}, \qquad s_0 = \sup(-S).$$

Prove the two assertions left open in the text:

$$(1) \quad -s_0 \le s \qquad \text{for every } s \in S,$$

$$(2) \quad t \le s \text{ for every } s \in S \qquad \Longrightarrow \quad t \le -s_0.$$

Step 1: justify that s_0 exists as a real number.

Since S is nonempty, choose $s_* \in S$. Then $-s_* \in -S$, so $-S$ is also nonempty. Since S is bounded below, there is a real number m with $m \le s$ for every $s \in S$. Negating this inequality reverses it, so $-s \le -m$ for every $s \in S$. Every element of $-S$ has the form $-s$ for such an s. Therefore $-m$ is an upper bound of $-S$. The completeness axiom applies and gives the real number $s_0 = \sup(-S)$.

Step 2: prove assertion (1), the lower-bound property.

Let $s \in S$ be arbitrary. By the definition of $-S$, its negative $-s$ belongs to $-S$. Because s_0 is an upper bound of that set,

$$-s \le s_0.$$

Negate both sides. The direction reverses, and $-(-s) = s$, giving

$$-s_0 \le s.$$

This holds for every $s \in S$, so $-s_0$ is a lower bound of S.

Step 3: prove assertion (2), the greatest-lower-bound property.

Let t be any real lower bound of S. This means $t \le s$ for every $s \in S$. Negating each of those inequalities gives

$$-s \le -t \qquad \text{for every } s \in S.$$

Hence $-t$ is an upper bound of $-S$. The least-upper-bound property of $s_0 = \sup(-S)$ now gives

$$s_0 \le -t.$$

Negating once more yields $t \le -s_0$. Thus every lower bound t of S is at most $-s_0$.

Conclusion. Assertion (1) says $-s_0$ is a lower bound, and assertion (2) says it is greater than or equal to every lower bound. Together they prove $\boxed{\inf S = -s_0 = -\sup(-S)}$. This establishes the lower-bound form of completeness from its upper-bound form.

4.10 Exercise 4.10: One natural number meeting two inequalities

What we must show. Given a real number $a > 0$, find one $n \in \mathbb{N}$ such that $1/n < a < n$.

Plan. It is not enough to find one integer for $1/n < a$ and a different integer for $a < n$. We choose a single integer larger than both a and $1/a$.

Step 1: form one positive real target.

Since $a > 0$, its inverse $1/a$ exists and is positive. Define

$$B = \max\{a, 1/a\}.$$

The number B is positive because both entries in this maximum are positive.

Step 2: apply the Archimedean property.

Use Theorem 4.6 with the positive increment 1 and the positive target B. It gives a natural number n satisfying

$$n \cdot 1 > B.$$

Thus $n > B \geq a$, so $a < n$, and $n > B \geq 1/a$, so $n > 1/a$.

Step 3: convert $n > 1/a$ into the reciprocal inequality.

Multiply $n > 1/a$ by the positive number a to get $na > 1$. Because $n \in \mathbb{N}$, we have $n > 0$, so division by n preserves strict inequality:

$$na > 1 \quad \Longrightarrow \quad a > \frac{1}{n}.$$

Conclusion. The same natural number n satisfies both comparisons: $\boxed{\frac{1}{n} < a < n}$. Both positivity checks are needed when multiplying and dividing inequalities.

4.11 Exercise 4.11: Every nonempty open interval contains infinitely many rationals

What we must show. If $a, b \in \mathbb{R}$ and $a < b$, use the denseness of $\mathbb{Q}$ to prove that there are infinitely many rational numbers r with $a < r < b$.

Step 1: define the set whose size is at issue.

Let

$$E = \{r \in \mathbb{Q} : a < r < b\}.$$

Theorem 4.7, applied to $a < b$, gives at least one rational in this interval. Thus E is nonempty. This first application alone does not prove infinitude; we must rule out the possibility that the set is finite.

Step 2: suppose there were only finitely many.

Assume, for a contradiction, that

$$E = \{r_1, r_2, \ldots, r_k\}$$

for some positive integer k. A nonempty finite set of real numbers has a maximum: compare its first two entries, keep the larger, and continue through the finitely many entries. Let

$$r_* = \max\{r_1, \ldots, r_k\}.$$

Since $r_* \in E$, it satisfies $a < r_* < b$.

Step 3: use density in a smaller interval.

The strict inequality $r_* < b$ permits another application of Theorem 4.7. There is a rational q with

$$r_* < q < b.$$

Because $a < r_* < q$, transitivity gives $a < q$. Together with $q < b$ and $q \in \mathbb{Q}$, this shows $q \in E$. But $q > r_*$, contradicting the choice of r_* as the largest member of E.

Conclusion. The set E is nonempty and cannot be finite. Therefore there are **infinitely many rational numbers between a and b**. The same density theorem can always be applied above the largest member of any proposed finite list.

4.12 Exercise 4.12: Irrational numbers are dense as well

What we must show. Let $I = \mathbb{R} \setminus \mathbb{Q}$. Prove that for any real numbers $a < b$ there is an $x \in I$ such that $a < x < b$.

Step 1: prove that a rational translate of $\sqrt{2}$ is irrational.

The irrationality of $\sqrt{2}$ was established in Section 2. Let $r \in \mathbb{Q}$ be arbitrary and suppose, for a contradiction, that $r + \sqrt{2}$ were rational. Write

$$r = \frac{m}{n}, \qquad r + \sqrt{2} = \frac{h}{k},$$

where $m, n, h, k \in \mathbb{Z}$, $n \neq 0$, and $k \neq 0$. Subtracting gives

$$\sqrt{2} = \frac{h}{k} - \frac{m}{n} = \frac{hn - mk}{kn}.$$

The numerator $hn - mk$ is an integer, the denominator kn is an integer, and $kn \neq 0$ because both k and n are nonzero. The last quotient would make $\sqrt{2}$ rational, a contradiction. Thus $r + \sqrt{2}$ is irrational for every $r \in \mathbb{Q}$, proving

$$\{r + \sqrt{2} : r \in \mathbb{Q}\} \subseteq I.$$

Step 2: shift the desired interval before choosing a rational.

Adding $-\sqrt{2}$ to $a < b$ preserves strict inequality:

$$a - \sqrt{2} < b - \sqrt{2}.$$

By Theorem 4.7, there is a rational number r satisfying

$$a - \sqrt{2} < r < b - \sqrt{2}.$$

The endpoints here are real numbers; they need not themselves be rational for the density theorem to apply.

Step 3: shift that rational back.

Define $x = r + \sqrt{2}$. Step 1 gives $x \in I$. Add $\sqrt{2}$ to all three expressions in the preceding two inequalities:

$$(a - \sqrt{2}) + \sqrt{2} < r + \sqrt{2} < (b - \sqrt{2}) + \sqrt{2}.$$

Associativity and additive cancellation simplify the endpoints, giving $a < x < b$.

Conclusion. The irrational number $\boxed{x = r + \sqrt{2}}$, where $a - \sqrt{2} < r < b - \sqrt{2}$ and $r \in \mathbb{Q}$, lies strictly between a and b. The initial shift is what ensures that adding $\sqrt{2}$ does not move our chosen point outside the desired interval.

4.13 Exercise 4.13: Three descriptions of an open interval

What we must show. For real a, b, c, prove the equivalence of

$$\text{(i)} \quad |a - b| < c,$$

$$\text{(ii)} \quad b - c < a < b + c,$$
$$\text{(iii)} \quad a \in (b - c, b + c).$$

We interpret (u, v) as the set $\{x \in \mathbb{R} : u < x < v\}$; this set is empty when $u \geq v$.

Step 1: prove that (i) implies (ii).

Assume $|a - b| < c$. Exercise 3.7(a), with bounded number $a - b$ and bound c, gives

$$-c < a - b < c.$$

Add b throughout. Strict inequalities are preserved by addition, so

$$-c + b < (a - b) + b < c + b.$$

Associativity and the additive inverse rule simplify the middle expression to a. Commutativity turns the endpoints into $b - c$ and $b + c$. We have obtained $b - c < a < b + c$, which is (ii).

Step 2: prove that (ii) implies (i).

Assume $b - c < a < b + c$. Add $-b$ throughout and simplify to obtain

$$-c < a - b < c.$$

Exercise 3.7(a) now gives $|a - b| < c$. This is (i).

Step 3: prove the equivalence of (ii) and (iii).

By the definition of the open interval,

$$(b - c, b + c) = \{x \in \mathbb{R} : b - c < x < b + c\}.$$

Membership of the particular number a in this set means exactly that $b - c < a < b + c$. Thus (ii) implies (iii), and (iii) implies (ii).

Step 4: verify explicitly what happens when $c \leq 0$.

If $c \leq 0$, statement (i) is false because every absolute value is nonnegative: it cannot satisfy $0 \leq |a - b| < c \leq 0$. Also $c \leq 0$ implies $c \leq -c$, so adding b gives $b + c \leq b - c$. Statement (ii) would imply $b - c < b + c$, contrary to that inequality. Thus (ii) is false. The interval in (iii) has its left endpoint at least as large as its right endpoint, so it is empty; statement (iii) is false too. This includes $c = 0$, when both endpoints equal b.

Conclusion. For $c > 0$, the steps above show that the three statements are equivalent; for $c \leq 0$, all three are false. Consequently, for every real a, b, c, $\boxed{\text{(i)} \iff \text{(ii)} \iff \text{(iii)}}$.

4.14 Exercise 4.14: Supremum and infimum of the sum of two sets

What we must show. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Define

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

Prove

$$\text{(a)} \quad \sup(A + B) = \sup A + \sup B, \qquad \text{(b)} \quad \inf(A + B) = \inf A + \inf B.$$

The notation $A + B$ means the set of all such sums, not a union.

Preliminary step: check nonemptiness, boundedness, and existence.

Since A and B are nonempty and bounded, their extrema as bounds exist as real numbers. Write

$$m_A = \inf A, \quad M_A = \sup A, \qquad m_B = \inf B, \quad M_B = \sup B.$$

Choose $a_0 \in A$ and $b_0 \in B$. Then $a_0 + b_0 \in A + B$, so the sum set is nonempty. For arbitrary $a \in A$ and $b \in B$, we have

$$m_A \le a \le M_A, \qquad m_B \le b \le M_B.$$

Adding the lower comparisons and then the upper comparisons gives

$$m_A + m_B \le a + m_B \le a + b \le M_A + b \le M_A + M_B.$$

Thus $A+B$ is bounded below by m_A+m_B and above by M_A+M_B. Completeness and its lower-bound corollary supply real numbers

$$m = \inf(A + B), \qquad M = \sup(A + B).$$

No assertion that an infimum or supremum belongs to A or B has been made.

Part (a), Step 1: prove the first inequality.

We just showed that every $a + b \in A + B$ satisfies $a + b \le M_A + M_B$. Hence $M_A + M_B$ is an upper bound of $A + B$. Its least upper bound is M, so

$$M \le M_A + M_B.$$

Part (a), Step 2: explain how to approach each separate supremum.

Let $\varepsilon > 0$. Because $\varepsilon/2 > 0$, $M_A - \varepsilon/2 < M_A$. The number $M_A - \varepsilon/2$ cannot be an upper bound of A: such an upper bound would be strictly smaller than the least upper bound M_A. Therefore there is an $a_\varepsilon \in A$ with

$$M_A - \frac{\varepsilon}{2} < a_\varepsilon \le M_A.$$

The same argument for B gives a $b_\varepsilon \in B$ with

$$M_B - \frac{\varepsilon}{2} < b_\varepsilon \le M_B.$$

These are approximation statements, not claims that $M_A \in A$ or $M_B \in B$.

Part (a), Step 3: prove the reverse inequality using a gap.

Suppose, for a contradiction, that $M < M_A + M_B$. Set

$$\varepsilon = M_A + M_B - M > 0.$$

Choose $a_\varepsilon, b_\varepsilon$ as in Step 2. Adding their two strict lower bounds gives

$$\begin{aligned} a_\varepsilon + b_\varepsilon &> \left(M_A - \frac{\varepsilon}{2}\right) + \left(M_B - \frac{\varepsilon}{2}\right), \\ &= M_A + M_B - \varepsilon, \end{aligned}$$

$$= M.$$

But $a_\varepsilon + b_\varepsilon \in A + B$, and M is an upper bound of this set, so the same sum must be at most M. This is a contradiction. Therefore $M_A + M_B \leq M$.

Conclusion. Both $M \leq M_A + M_B$ and $M_A + M_B \leq M$ hold. By antisymmetry,

$$\boxed{\sup(A + B) = \sup A + \sup B}.$$

Splitting the positive gap into two halves leaves enough room to choose one near-maximal element from each set.

Part (b), Step 1: prove the first inequality.

The preliminary step showed $m_A + m_B \leq a + b$ for every $a + b \in A + B$. Thus $m_A + m_B$ is a lower bound of the sum set. Since m is the greatest lower bound,

$$m_A + m_B \leq m.$$

Part (b), Step 2: explain how to approach the separate infima.

Let $\varepsilon > 0$. The number $m_A + \varepsilon/2$ is strictly larger than m_A. It cannot be a lower bound of A, since m_A is its greatest lower bound. Therefore there is an $a_\varepsilon \in A$ with

$$m_A \leq a_\varepsilon < m_A + \frac{\varepsilon}{2}.$$

Likewise there is a $b_\varepsilon \in B$ with

$$m_B \leq b_\varepsilon < m_B + \frac{\varepsilon}{2}.$$

Again no attainment of either infimum is being assumed.

Part (b), Step 3: prove the reverse inequality using a gap.

Suppose, for a contradiction, that $m > m_A + m_B$. Put

$$\varepsilon = m - m_A - m_B > 0.$$

Choose $a_\varepsilon, b_\varepsilon$ from Step 2. Adding their two strict upper bounds gives

$$\begin{aligned} a_\varepsilon + b_\varepsilon &< \left(m_A + \frac{\varepsilon}{2}\right) + \left(m_B + \frac{\varepsilon}{2}\right), \\ &= m_A + m_B + \varepsilon, \\ &= m. \end{aligned}$$

This sum belongs to $A + B$. But m is a lower bound, so every member of that set must be at least m. The contradiction rules out $m > m_A + m_B$, giving $m \leq m_A + m_B$.

Conclusion. Combining the two comparisons gives $\boxed{\inf(A + B) = \inf A + \inf B}$. The argument applies whether or not the extremal bounds of the original sets are attained.

4.15 Exercise 4.15: Comparison using reciprocal natural-number errors

What we must show. Suppose $a, b \in \mathbb{R}$ and

$$a \leq b + \frac{1}{n} \qquad \text{for every } n \in \mathbb{N}.$$

Prove $a \leq b$.

Plan. If $a > b$, the positive gap $a - b$ can be made larger than one of the numbers $1/n$. That particular n contradicts the hypothesis.

Step 1: assume the opposite of the desired conclusion.

Suppose $a \leq b$ is false. Total ordering then gives $a > b$, so

$$d = a - b > 0.$$

Step 2: obtain a strict reciprocal estimate from the Archimedean property.

Apply Theorem 4.6 to the positive numbers d and 1. There is an $n \in \mathbb{N}$ with $nd > 1$. Since $n > 0$, division by n gives

$$d > \frac{1}{n}.$$

The strict inequality is needed: an estimate $1/n \leq d$ alone would not exclude equality at the next step.

Step 3: compare with the hypothesis for that particular n.

Substitute $d = a - b$ and add b to get

$$\frac{1}{n} < a - b \quad \Longrightarrow \quad b + \frac{1}{n} < a.$$

The hypothesis applies to every natural number, so it applies to this one and gives $a \leq b + 1/n$. These inequalities cannot both hold: they would give $a \leq b + 1/n < a$.

Conclusion. The contradiction proves $\boxed{a \leq b}$. Exercise 3.8 tested a against every real number above b. Here only the particular numbers $b + 1/n$ are tested; the Archimedean property guarantees that at least one of them detects any hypothetical positive gap $a - b$.

4.16 Exercise 4.16: A real number as the supremum of the rationals below it

What we must show. For each $a \in \mathbb{R}$, prove

$$\sup\{r \in \mathbb{Q} : r < a\} = a.$$

Step 1: define the set and prove that it is nonempty.

Fix an arbitrary real a and let

$$S_a = \{r \in \mathbb{Q} : r < a\}.$$

The inequality $a - 1 < a$ is strict because $1 > 0$. Apply the denseness of $\mathbb{Q}$ to the interval $(a - 1, a)$. It gives a rational r_0 with

$$a - 1 < r_0 < a.$$

Thus $r_0 \in S_a$, proving nonemptiness. This construction works even for $a \leq 0$; one must not assume that 0 belongs to S_a.

Step 2: prove that a is an upper bound.

By definition, every $r \in S_a$ satisfies $r < a$, and therefore $r \leq a$. Hence a is an upper bound of S_a. In particular, S_a is nonempty and bounded above, so its real supremum exists by completeness.

Step 3: prove that every smaller candidate fails to be an upper bound.

Take any real $u < a$. Denseness of $\mathbb{Q}$ gives a rational r with

$$u < r < a.$$

Since $r \in \mathbb{Q}$ and $r < a$, this r belongs to S_a. The additional inequality $u < r$ shows that u is not an upper bound of S_a: it is exceeded by the specific element $r \in S_a$.

Step 4: express the least-upper-bound conclusion.

Step 2 showed that a is an upper bound. Step 3 showed that every number strictly less than a fails to be an upper bound. Equivalently, if U is any upper bound, $U < a$ is impossible, and the real order therefore forces $a \leq U$. Thus a is at most every upper bound while being one itself.

Conclusion. The two defining obligations for a supremum are satisfied, so $\boxed{\sup\{r \in \mathbb{Q} : r < a\} = a}$ for every real a. The number a need not be rational, and it is excluded from the set even when it is rational; the supremum need not be attained.